CSCE 314
Programming Languages

Functors, Applicatives, and Monads

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Motivation – Generic Functions

A common programming pattern can be abstracted out as a definition.

For example:

```
inc :: [Int] -> [Int]
inc []     = []
inc (n:ns) = n+1 : inc ns

sqr :: [Int] -> [Int]
sqr []     = []
sqr (n:ns) = n^2 : sqr ns
```

Both functions are defined in the same manner!

```
inc :: [Int] -> [Int]
inc []     = []
inc (n:ns) = n+1 : inc ns

sqr :: [Int] -> [Int]
sqr []     = []
sqr (n:ns) = n^2 : sqr ns
```

```
inc = map (+1)
```

```
sqr = map (^2)
```

Using `map`

```
map :: (a -> b) -> [a] -> [b]
map f []     = []
map f (n:ns) = f n : map f ns
```

Functors

Class of types that support mapping of function. For example, lists and trees.

```
class Functor f where
  fmap :: (a -> b) -> f a -> f b
```

(f a) is a data structure that contains elements of type a

fmap takes a function of type (a->b) and a structure of type (f a), applies the function to each element of the structure, and returns a structure of type (f b).

Functor instance example 1: the list structure []

```
instance Functor [] where
  -- fmap :: (a -> b) -> [a] -> [b]
  fmap = map
```

Functor instance example 2: the Maybe type

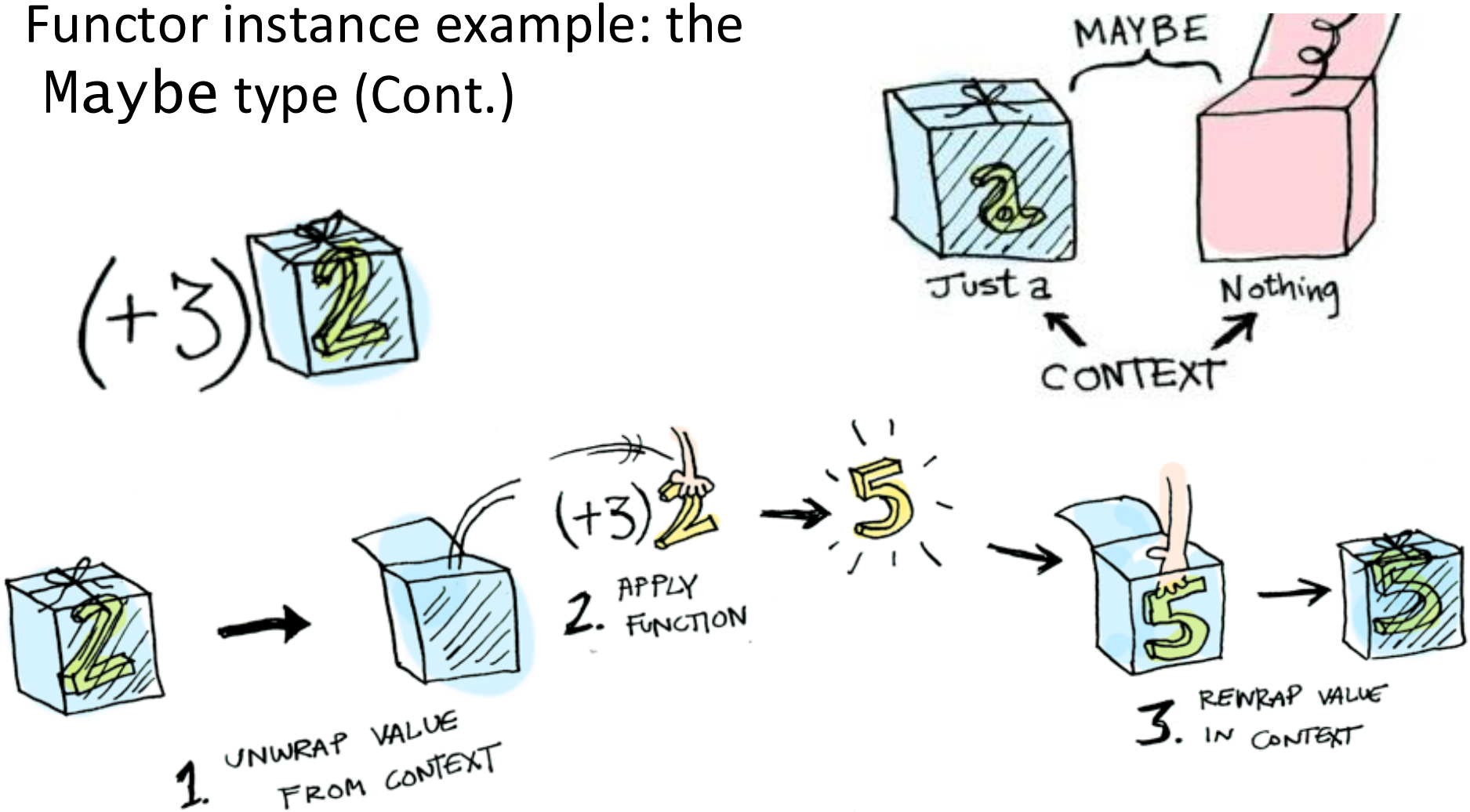
```
data Maybe a = Nothing | Just a
```

```
instance Functor Maybe where
  -- fmap :: (a -> b) -> Maybe a -> Maybe b
  fmap _ Nothing  = Nothing
  fmap g (Just x) = Just (g x)
```

Now, you can do

```
> fmap (+1) Nothing
Nothing
> fmap not (Just True)
Just False
```

Functor instance example: the Maybe type (Cont.)



Picture source:
http://adit.io/posts/2013-04-17-functors,_applicatives,_and_monads_in_pictures.html

Functor instance example 3: the Tree type

```
data Tree a = Leaf a | Node (Tree a) (Tree a)
              deriving Show
```

```
instance Functor Tree where
  -- fmap :: (a -> b) -> Tree a -> Tree b
  fmap g (Leaf x)   = Leaf (g x)
  fmap g (Node l r) = Node (fmap g l) (fmap g r)
```

Now, you can do

```
> fmap (+1) (Node (Leaf 1) (Leaf 2))
Node (Leaf 2) (Leaf 3)
> fmap (even) (Node (Leaf 1) (Leaf 2))
Node (Leaf False) (Leaf True)
```

Functor laws

```
1. fmap id       = id
2. fmap (g . h) = fmap g . fmap h
```

1. fmap preserves the identity function

2. fmap also preserves the function composition, where g has type b -> c and h has type a -> b

3. The functor laws ensure that fmap does perform a mapping operation, without altering the natural property of the data structure.

Benefits of Functors

1. `fmap` can be used to process the elements of any structure that is functorial.
2. Allows us to define generic functions that can be used with any functor.

Example: increment (`inc`) function can be used with any functor with `Int` type elements

```
inc :: Functor f => f Int -> f Int
inc = fmap (+1)
> inc (Just 1)
Just 2
> inc [1,2,3]
[2,3,4]
> inc (Node (Leaf 1) (Leaf 2))
Node (Leaf 2) (Leaf 3)
```

Want to be more flexible?

Functors abstract the idea of mapping a function over each element of a structure.

```
class Functor f where
    fmap :: (a -> b) -> f a -> f b
```

Only one argument function!

The first argument of `fmap` is a function that takes one argument, but we want more flexibility! We want to be able to use functions that take any number of arguments.

```
class Functor f => Applicative f where
   pure  :: a -> f a
   (<*>) :: f (a -> b) -> f a -> f b
```

Applicative

```
class (Functor f) => Applicative f where
   pure  :: a -> f a
   (<*>) :: f (a -> b) -> f a -> f b
```

The function `pure` takes a value of any type as its argument and returns a structure of type `f a`, that is, an applicative functor that contains the value.

The operator <*> is a generalization of function application for which the argument function, the argument value, and the result value are all contained in `f` structure.

```
fmap g x = pure g <*> x = g <$> x -- infix version of fmap
```

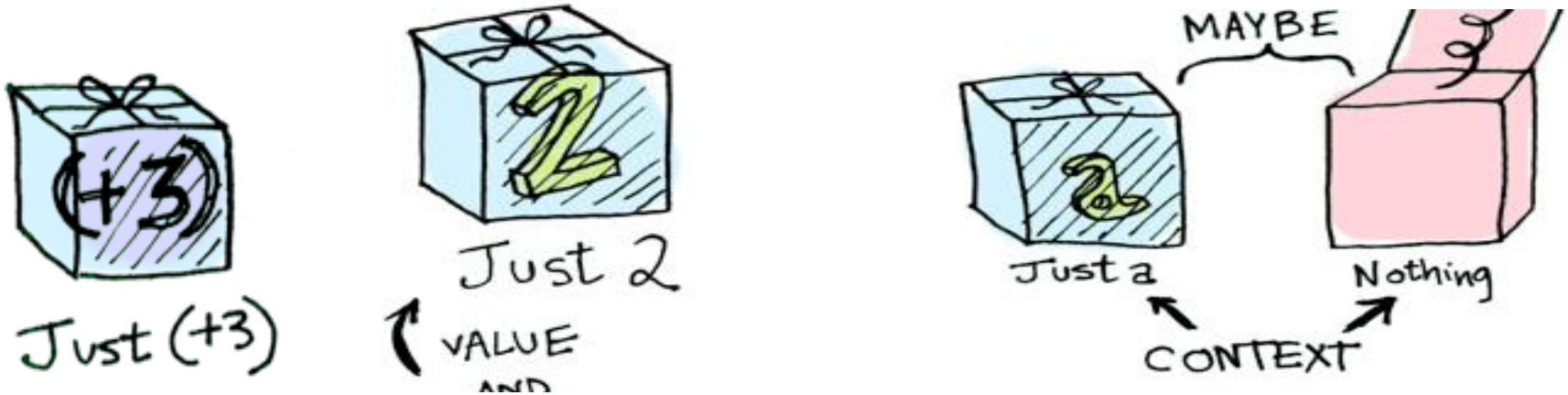
Applicative functor instance example 1: Maybe

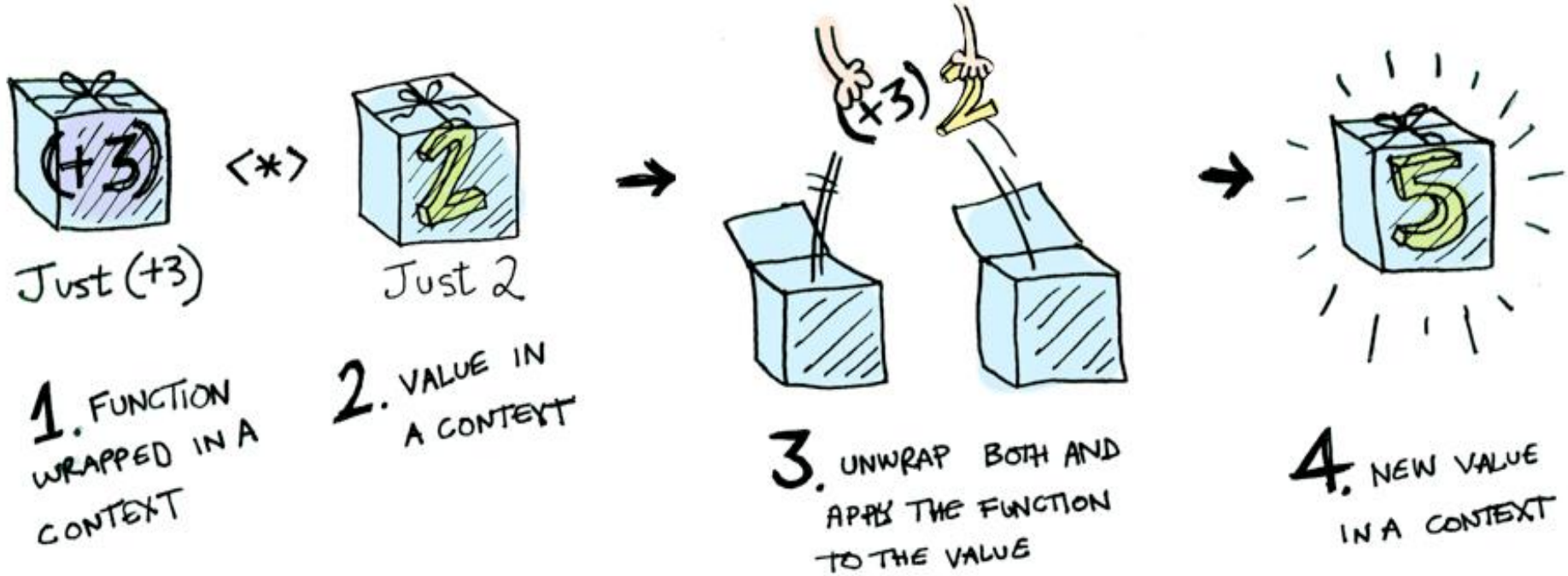
```
data Maybe a = Nothing | Just a
```

```
instance Applicative Maybe where
  -- pure :: a -> Maybe a
  pure = Just
  -- (<*>) :: Maybe (a->b) -> Maybe a -> Maybe b
  Nothing  <*> _  = Nothing
  (Just g) <*> mx = fmap g mx
```

```
> pure (+1) <*> Nothing
Nothing
> pure (+) <*> Just 2 <*> Just 3
Just 5
> mult3 x y z = x*y*z
> pure mult3 <*> Just 1 <*> Just 2 <*> Just 4
Just 8
```

Applicative functor instance example: Maybe (Cont.)





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Applicative functor instance example 2: list type []

```
instance Applicative [] where
  -- pure :: a -> [a]
  pure x = [x]
  -- (<*>) :: [a -> b] -> [a] -> [b]
  gs <*> xs  = [ g x | g <- gs, x <- xs ]
```

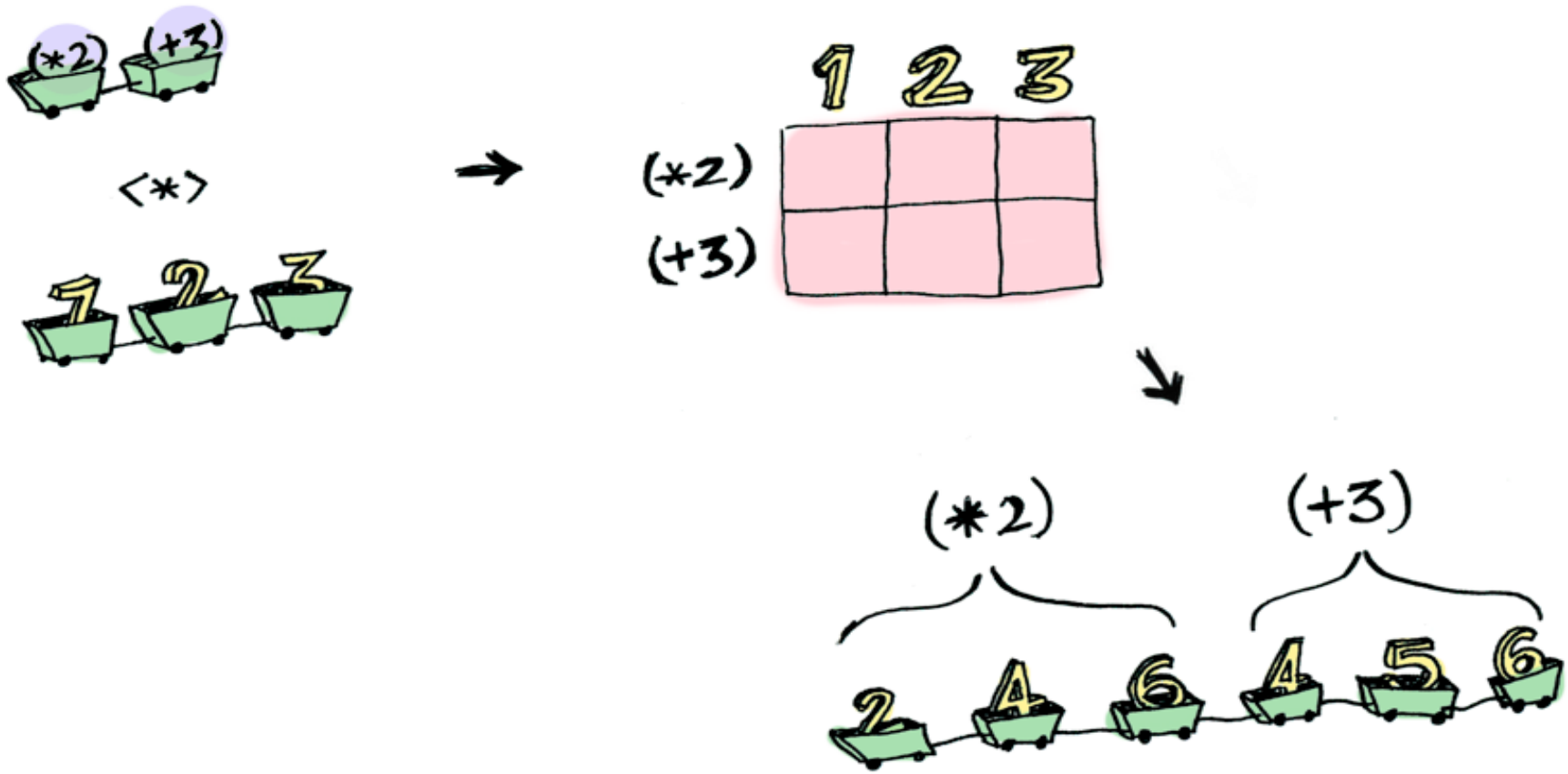
`pure` transforms a value into a singleton list.

<*> takes a list of functions and a list of arguments, and applies each function to each argument in turn, returning all the results in a list.

```
> pure (+1) <*> [1,2,3]
[2,3,4]
> pure (+) <*> [1,3] <*> [2,5]
[3,6,5,8]
> pure (:) <*> "ab" <*> ["cd","ef"]
["acd","aef","bcd","bef"]
```

Applicative functor instance example: [] (Cont.)

```
> [(*2), (+3)] <*> [1,2,3]
[2,4,6,4,5,6]
```



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Applicative functor instance example 3: the Tree type

```
data Tree a = Leaf a | Node (Tree a) (Tree a) deriving Show
```

```
instance Applicative Tree where
  -- pure :: a -> Tree a
  pure = Leaf
  -- (<*>) :: Tree (a -> b) -> Tree a -> Tree b
  (Leaf g) <*> mx = fmap g mx
  (Node l r) <*> mx = Node (l <*> mx) (r <*> mx)
```

```
tree1 = Node (Leaf 1) (Leaf 2)
tree2 = Node (Leaf 10) (Leaf 20)
```

```
> pure (+1) <*> tree1
Node (Leaf 2) (Leaf 3)
> pure (+) <*> tree1 <*> tree2
Node (Node (Leaf 11) (Leaf 21)) (Node (Leaf 12) (Leaf 22))
```

Applicative laws

```
pure id <*> x    = x
pure (g x)       = pure g <*> pure x
x <*> pure y     = pure (\g -> g y) <*> x
x <*> (y <*> z) = (pure (.) <*> x <*> y) <*> z
```

1. pure preserves the identity function
2. pure also preserves function application
3. When an effectful function is applied to a pure argument, the order in which the two components are evaluated does not matter.
4. The operator <*> is associative (modulo types that are involved).

Monads

```
class (Applicative m) => Monad m where
   return  :: a -> m a
   (>>=) :: m a -> (a -> m b) -> m b
   return = pure
```

- Roughly, a monad is a strategy for combining computations into more complex computations.
- Another pattern of *effectful programming* (applying pure functions to (side-)effectful arguments)
- `(>>=)` is called "bind" operator.
- Note: `return` may be removed from the Monad class in the future and become a library function instead.

Monad instance example 1: Maybe

```
data Maybe a = Nothing | Just a
```

```
instance Monad Maybe where
 -- (>>=) :: Maybe a -> (a -> Maybe b) -> Maybe b
 Nothing  >>= _ = Nothing
 (Just x) >>= f = f x
```

```
div2 x = if even x then Just (x `div` 2) else Nothing
```

```
> (Just 10) >>= div2
Just 5
> (Just 10) >>= div2 >>= div2
Nothing
> (Just 10) >>= div2 >>= div2 >>= div2
Nothing
```

Monad instance example 2: list type []

```
instance Monad [] where
  -- (>>=) :: [a] -> (a -> [b]) -> [b]
  xs >>= f = [y | x <- xs, y <- f x]
```

```
pairs :: [a] -> [b] -> [(a,b)]
pairs xs ys = do x <- xs
                 y <- ys
                 return (x,y)
```

```
pairs xs ys = xs >>= \x ->
              ys >>= \y ->
              return (x,y)
```

```
> pairs [1,2] [3,4]
[(1,3),(1,4),(2,3),(2,4)]
```

Monad instance example 3: the Tree type

```
data Tree a = Leaf a | Node (Tree a) (Tree a) deriving Show
```

```
instance Monad Tree where
  -- (>>=) :: Tree a -> (a -> Tree b) -> Tree b
  (Leaf x) >>= f = f x
  (Node l r) >>= f = Node (l >>= f) (r >>= f)
```

```
tree1 = Node (Leaf 1) (Leaf 2)
tree2 = Node (Leaf 10) (Leaf 20)
feven = \x -> Leaf (even x)
fnot  = \x -> Leaf (not x)
```

```
> tree1 >>= feven >>= fnot
Node (Leaf True) (Leaf False)
> tree2 >>= feven
Node (Leaf True) (Leaf True)
```

Monad laws

```
return x >>= f     =  f x -- left identity
mx >>= return      =  mx  -- right identity
(mx >>= f) >>= g   =  mx >>= (\x -> (f x >>= g))
```

1. If we return a value and then feed it into a monadic function, this should give the same result as simply applying the function to the value.
2. If we feed the result of a monadic computation into the function return, this should give the same result as simply performing the computation.
3. >>= is associative